

Computer Fundamentals

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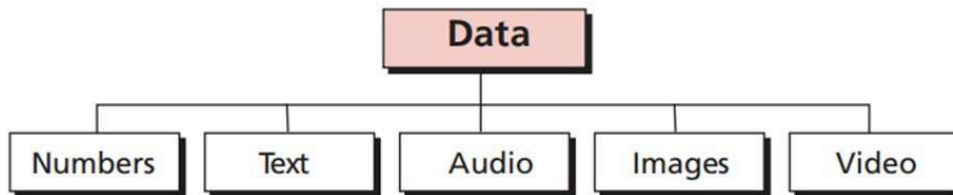
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Chapter2 Data representation and operation

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DATA TYPES

- **Data today come in different forms including numbers, text, audio, images, and video.**



The computer industry uses the term 'multimedia' to define information that contains numbers, text, audio, images, and video.

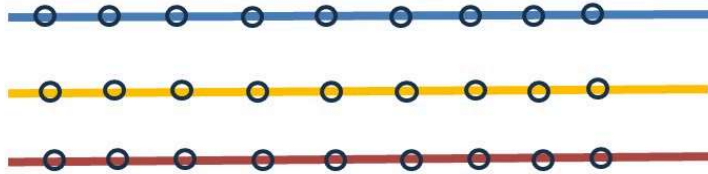
1. Number Systems

- **Number systems**
 - The decimal system (base 10)
 - The hexadecimal system (base 16)
 - The binary system (base 2)
 - The octal system (base 8)

1. Number Systems

- **The Decimal system**

- [review about decimal](#)



- Symbols(digits): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- eg: 18, 1501,
- Digits: the symbols in the system.
- Base: the number of symbols used in the system.

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1. Number Systems

- **The Hexadecimal system**

- [review about hexadecimal](#)

- 半斤八两 half Jin = 8 Liang
- 1 Jin = 16 Liang

- Symbols(digits): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- eg: F, 10F

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1. Number Systems

- **Position number systems:**

System	Base	Symbols(digits)	Eg
Decimal	10	0,1,2,3,4,5,6,7,8,9	234.56
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F	(AB.1F) ₁₆
Binary	2	0,1	(101.11) ₂
Octal	8	0,1,2,3,4,5,6,7	(163.25) ₈

- $9D + 1D = 10D$, $FH + 1H = 10H$, $(7)8 + (1)8 = (10)8$, $1B + 1B = 10B$
- $393 = 300 + 90 + 3$
- $= 3 \times 100 + 9 \times 10 + 3 \times 1$
- $= 3 \times 10^2 + 9 \times 10^1 + 3 \times 10^0$ (2,1,0 represent the power of base, and the power corresponds the position of digits)

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1. Number Systems

- **Position number systems(Positional Notation):**

- A system of expressing numbers in which the digits are arranged in succession, the position of each digit has a place value, and the number is equal to the sum of the products of each digit by its place value.
- In a positional number system, the position a symbol occupies in the number determines the value it represents.

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1. Number Systems

Base: number of symbols used in the system. (R=2,8,16,10)

Digits: the symbols in the R system. (R=2,8,16,10)

Place/position of digit: the first digit to the left of the radix point is the 0th place.

Position corresponds the power of digit.

Integer part: the further to the left the digits is, the larger the value.

Fractional part: the further to the right the digit is, the smaller the value.

Place value: different position determines the different value.

1. Number Systems

eg: $393.124 = 3 \cdot 10^2 + 9 \cdot 10^1 + 3 \cdot 10^0 + 1 \cdot 10^{-1} + 2 \cdot 10^{-2} + 4 \cdot 10^{-3}$

digit	3	9	3	1	2	4
position	2	1	0	-1	-2	-3
power	2	1	0	-1	-2	-3
weight	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}

1. Number Systems

- **Convert the base R to base 10:**
 - Determine the value of a base R system number
 - Eg: base 16 number (19A.D)H
 - Eg: 14H=(?)D
 - Eg: (257)₈=(?)D
 - Eg: 101011.101B=(?)D

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2. Conversion

- **Summary of the four positional systems**
 - summary and examples

System	Base	Symbols	Examples
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	2345.56
Binary	2	0, 1	(1001.11) ₂
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	(156.23) ₈
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	(A2C.A1) ₁₆

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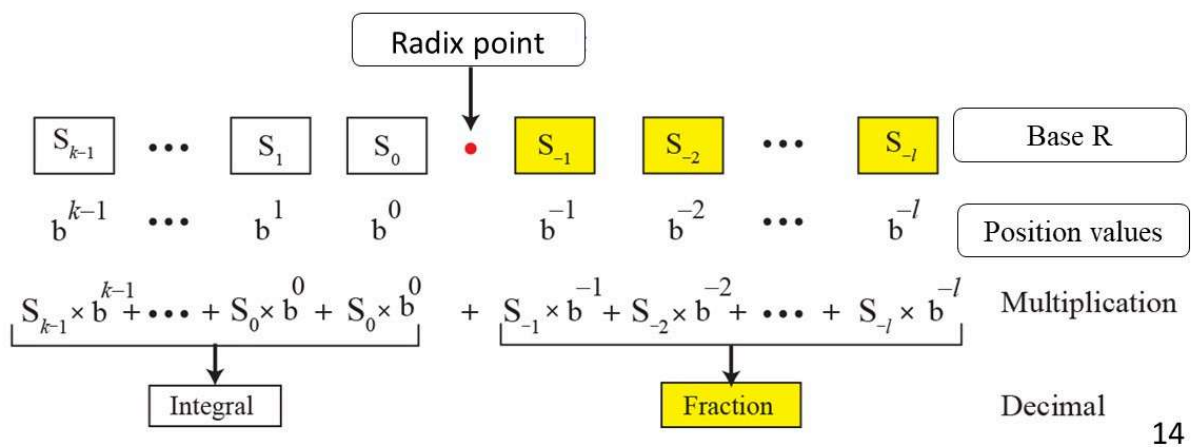
2. Conversion

- **Summary of the four positional systems**
 - comparison of numbers in the four systems

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	13 F

2. Conversion

- **Any base to decimal conversion**
 - solution



2. Conversion

- Any base to decimal conversion
 - examples

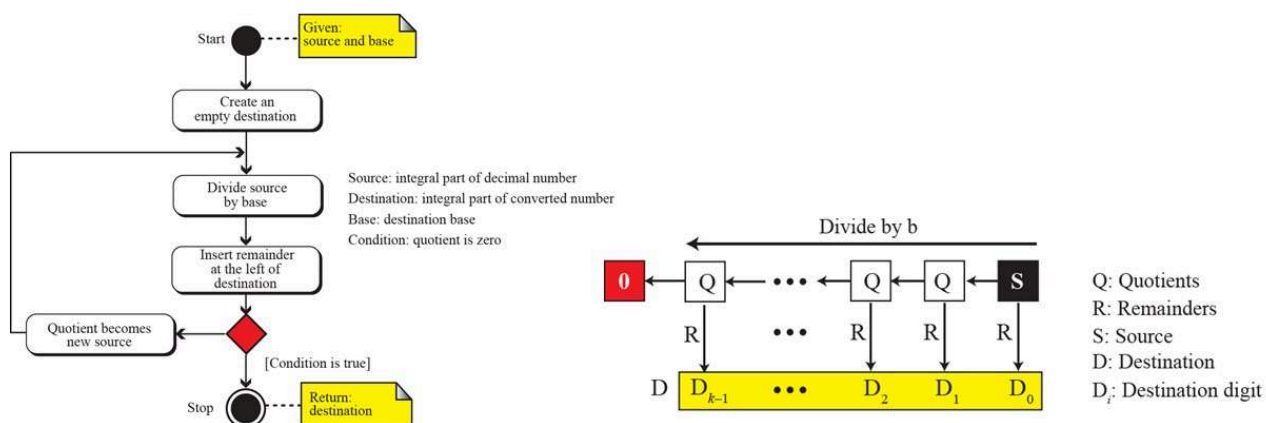
19A.DH=(?)D

1011.101B=(?)D

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2. Conversion

- Decimal to any base
 - (1) Converting the integral part



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2. Conversion

- **Converting the integral part**
 - examples(binary): addition method

position	7	6	5	4	3	2	1	0	-1	-2	-3
	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
value	128	64	32	16	8	4	2	1	0.5	0.25	0.125

9=(?)B

98=(?)B

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2. Conversion

- **Converting the integral part**
 - examples (binary): division method

98=(1100010)B

2	98		remainder	
2	49	-----	0	↑ Low position
2	24	-----	1	
2	12	-----	0	
2	6	-----	0	
2	3	-----	0	
2	1	-----	1	
	0	-----	1	High position

$$\begin{aligned}
 98 &= 2 * 49 + 0 \\
 &= 2 * (2 * 24 + 1) + 0 \\
 &= 2 * (2 * (2 * 12 + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * 6 + 0) + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * (2 * 3 + 0) + 0) + 0) + 1) + 0 \\
 &= 2 * (2 * (2 * (2 * (2 * 1 + 1) + 0) + 0) + 0) + 1) + 0
 \end{aligned}$$

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2. Conversion

- **Converting the integral part**
 - Examples(hexadecimal)

479=(?)H

254=(?)H

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2. Conversion

- **Converting the integral part**
 - Examples(octal)

25=(?)8

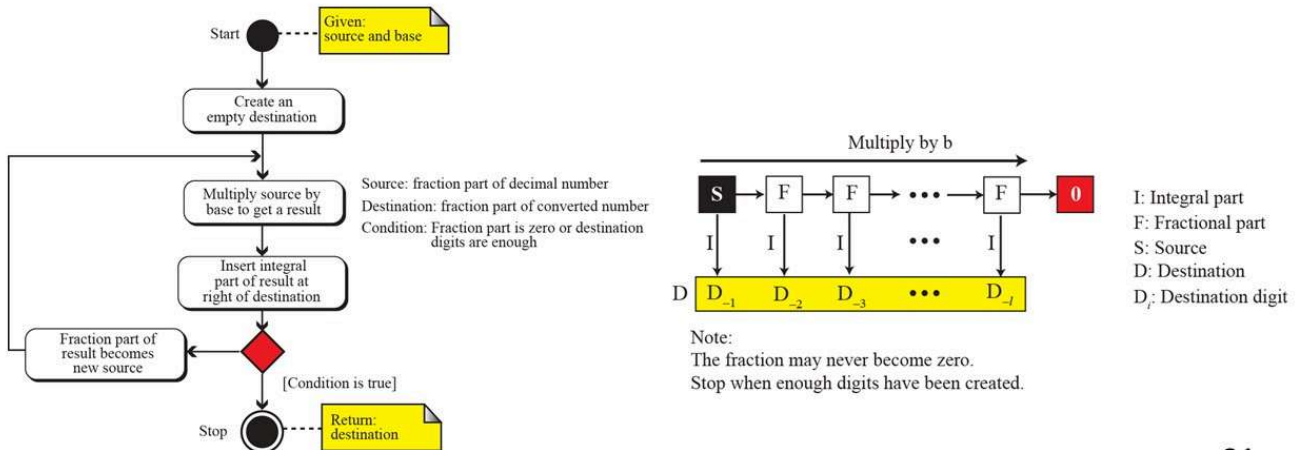
112=(?)8

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2. Conversion

- **Decimal to any base**

- **(2) Converting the fractional part**



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2. Conversion

- **Converting the fractional part**

- **examples (binary): multiplication method**

$$0.125 = (?)_B$$

$$0.2 = (?)_B$$

Some decimal fractions cannot be accurately converted to binary.
 Decimal to binary conversion could lead to an infinite digit sequence.

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2. Conversion

- **Converting the fractional part**
 - examples (hexadecimal): multiplication method

$$0.625 = (?)_{\text{H}}$$

$$0.4 = (?)_{\text{H}}$$

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2. Conversion

- **Converting the fractional part**
 - examples (octal): multiplication method

$$0.625 = (?)_8$$

$$0.4 = (?)_8$$

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2. Conversion

$$\underline{25.625}_{10} = (19.A)_{16}$$

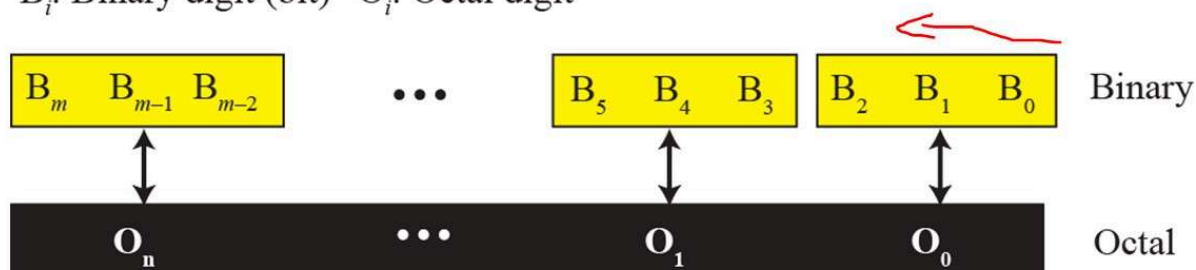
$$\underline{25.625}_{10} = (31.5)_8$$

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2. Conversion

- **binary-octal conversion**
 - three bits is one octal digit.

B_i : Binary digit (bit) O_i : Octal digit



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2. Conversion

- **binary-octal conversion**

- three bits is one octal digit.

$$\begin{array}{r} 010110101 \mid 101010B = (\quad)_8 \\ \underline{2 \quad 6 \quad 5} \quad \underline{5 \quad 2} \end{array} \xrightarrow{\text{red}} 265.52$$

$$\begin{array}{r} 001111.110 = (\quad)_8 \mid 7.6 \\ \underline{1 \quad 7 \quad 0} \end{array}$$

$$(123.25)_8 = (1010011.01)_B$$

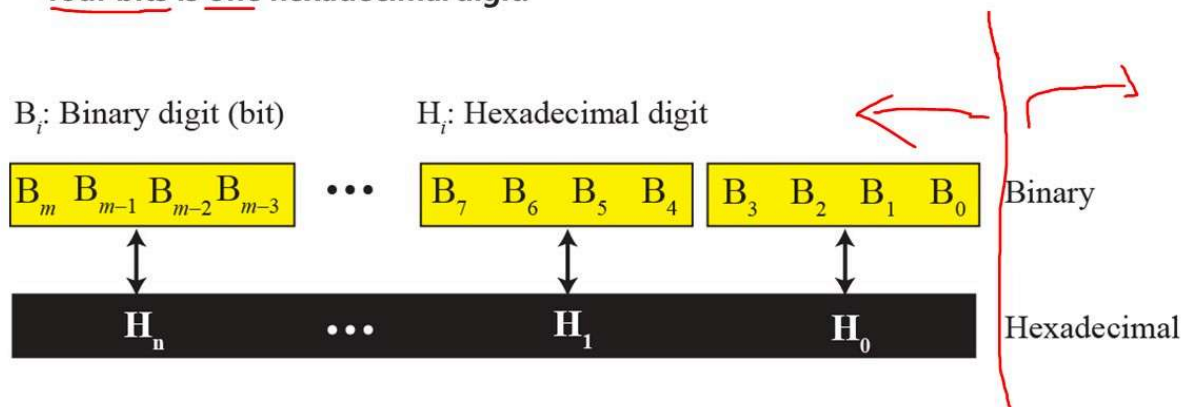
$$(26.32)_8 = (10110.01101)_B$$

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2. Conversion

- **binary-hexadecimal conversion**

- four bits is one hexadecimal digit.



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2. Conversion

1 bit 4 bits

- **binary-hexadecimal conversion**
 - four bits is one hexadecimal digit.

$1011010110101000B = ()H$ B5.A8
 $00011001.101B = ()H$ 19.A

38.D2H = (111000.1101001)B

16.EH = (0001 0110.1110)B

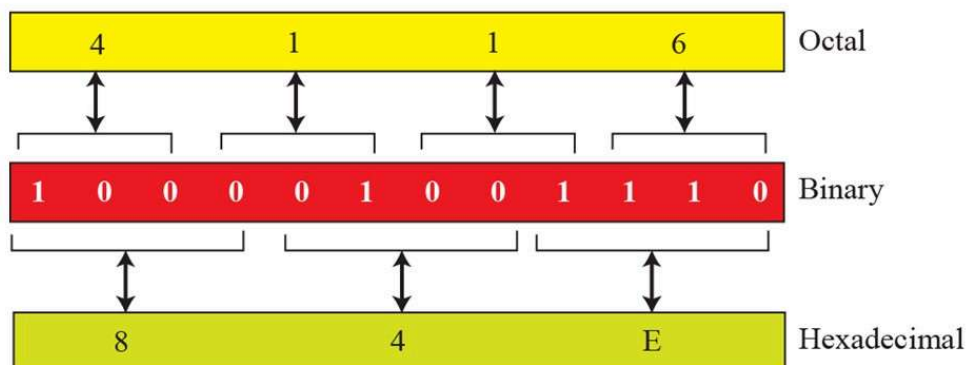
= (10110.111)B

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2. Conversion

8 $\leftrightarrow$ 6

- **octal-hexadecimal conversion**
 - We first convert the number to binary, then we rearrange the bits in groups of three/four bits to find the octal/hexadecimal equivalent.



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2. Conversion

- octal-hexadecimal conversion

$$(4116.1)_8 = (\quad)_{16}$$

$$\begin{aligned}(4116.1)_8 &= (\underline{100} \ \underline{001} \ \underline{001} \ \underline{110.001})_B \quad 8 \rightarrow 2 \\ &= (1000 \ 0100 \ 1110.0010)_B \\ &= (8 \ 4 \ E \ .2)_{16} \\ &= (84E.2)_{16}\end{aligned}$$

$$(AC.E)_{16} = (\quad)_8$$

Decimal	Binary	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

单选题 1分

1. Which number shouldn't be octal.

- A 123
- B 1011.101
- C 28.6
- D 642

填空题 1分

2. Convert the number 110D to binary [填空1]

填空题 1分

3. Convert the number 12.625D to binary [填空1]

填空题 1分

4. Convert the number 14.75D to octal [填空1]

填空题 1分

5. Convert the number 21.375D to hexadecimal [填空1]

填空题 1分

6. Convert the number 11001.110B to hexadecimal [填空1]

填空题 1分

7. Convert the number 11001.110B to octal [填空1]

填空题 1分

8. Convert the number A8.1H to octal [填空1]